

Problem Set 1

Due on Thursday, August 5 at 11:59 PM (if you want feedback) [It is fine to hand solutions to selected problems.]

1. Find the indicated limit, or explain why it does not exist.

(a) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x)$
(b) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$
(c) $\lim_{x \rightarrow 0^+} \sqrt{x} \ln x$
(d) $\lim_{x \rightarrow \infty} x[\ln(x + 5) - \ln x]$
(e) $\lim_{x \rightarrow \infty} \left(1 - \frac{2}{x} + \frac{7}{x^2}\right)^x$
(f) $\lim_{x \rightarrow 0} \frac{e^{3x} - 1 - 3x}{x^2}$

2. Find $f'(x)$ for

(a) $f(x) = \ln(x^4 + 1), \quad x \in \mathbb{R}$
(b) $f(x) = x^3 \sin(2x) + \cos(e^{x^2}), \quad x \in \mathbb{R}$
(c) $f(x) = \frac{(\sqrt{x} + 3)e^{-x}}{x^6 + 1}, \quad x > 0$
(d) $f(x) = (1 + x^2)^{3x}$

3. Assume that $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are twice differentiable and that

$$\begin{aligned} f(0) &= 5, & f(1) &= 3, & g(0) &= 1, & g(1) &= 2, \\ f'(0) &= -1, & f'(1) &= 4, & g'(0) &= 6, & g'(1) &= -3, \\ f''(0) &= -2, & f''(1) &= 5, & g''(0) &= 2, & g''(1) &= -3. \end{aligned}$$

Define $F, G : \mathbb{R} \rightarrow \mathbb{R}$ by

$$F(x) = f(x)g(x), \quad G(x) = f(g(x)), \quad x \in \mathbb{R}.$$

Determine $F''(1)$ and $G''(0)$.

4. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = x^3 e^{-x^5}, \quad x \in \mathbb{R}.$$

Determine $f^{(43)}(0)$, i.e. the value at $x = 0$ of the 43rd derivative of $f(x)$. (There is a quick way to do this.)

5. An investment banker can swim at the speed of 3 feet per second and run at the speed of 5 feet per second. She is in the water at a distance of 1000 feet from the nearest point A on a long straight shoreline. She spots a client sitting on the shoreline 2000 feet from point A . She wants to get to the client in the shortest possible time by swimming to a point on the shore between A and the client, and then running the rest of the way to the client. How far away from point A should she land on the shore?

6. Find the Maclaurin series (i.e., the Taylor series about $x_0 = 0$) for the function f defined by

(a) $f(x) = \ln(1 + x), \quad x \in (-1, 1),$

(b) $f(x) = \ln(1 + x^2), \quad x \in (-1, 1).$

7. Evaluate each of the following indefinite integrals.

(a) $\int (\sqrt{x} + 1)^2 dx$

(b) $\int e^{\sqrt{x}} dx$

(c) $\int x^4 (\ln x)^2 dx$

(d) $\int \frac{x^2}{\sqrt{x+1}} dx$

(e) $\int x^3 \cos(x^2) dx$

8. Evaluate each of the following definite integrals.

(a) $\int_1^3 \frac{x}{\sqrt{x+1}} dx$

(b) $\int_{-1}^1 \sqrt{x^2 + x^4} dx$

(c) $\int_0^{\ln 2} \frac{e^x + 5}{e^{4x}} dx$

9. Given that $\int_0^1 f(x) dx = 5$, $\int_0^2 f(x) dx = 3$, and $\int_2^3 f(x) dx = 5$, find $\int_1^3 f(x) dx$.

10. Suppose that f is differentiable and satisfies $f(1) = 0$ and

$$f'(x) = e^{-x^2} \quad \text{for all } x \in [0, 1].$$

Evaluate $\int_0^1 f(x) dx$.

11. Find $G'(x)$ for

(a) $G(x) = \int_x^5 e^{t^3} dt$

(b) $G(x) = \int_{\ln x}^{\sqrt{x}} \sqrt{1+t^2} dt$

12. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for every $x \in \mathbb{R}$ and every positive integer n we have

$$n^2 \int_x^{x+\frac{1}{n}} f(t) dt = nf(x) + 1.$$

Justify your conclusion completely.